

JOHN SEBASTIAN

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Research Interests

Fluid Mechanics; Applied Mathematics; Soft Matter; Biophysics

Education

Technical University of Denmark (DTU) 2023 - 2026

Doctor of Philosophy (Physics)

Soft Matter and Biophysics Group ([Jensen Research](#))

Thesis: Physical Constraints on Biological Form and Function

Ben-Gurion University of the Negev, Israel (BGU) 2021 - 2023

Master of Science (Mechanical Engineering)

Fluid Mechanics Laboratory ([Green Lab](#))

Thesis: Electrical Circuit Modelling of Nanofluidic Systems

College of Engineering, Trivandrum (CET)

University of Kerala, India

2013 - 2017

Bachelor of Technology (Mechanical Engineering)

Micro/Nanofluidics Laboratory ([RS Kumar Lab](#))

Thesis: Inverse Design of Short Span Hydrofoils

Publications

3. **J Sebastian** and K H Jensen. “The geometry of Nature’s stingers is universal due to stochastic mechanical wear” [Proc. Natl. Acad. Sci. U.S.A. 123 \(10\) \(2026\)](#)
2. **J Sebastian**, A L Schødt, and K H Jensen. “Experiments on a sphere settling towards a boundary in a viscous liquid under the influence of a magnetic force” [Journal of Fluid Mechanics \(2025\)](#)
1. **J Sebastian** and Y Green. “Electrical Circuit Modelling of Nanofluidic Systems”
(*On journal cover*) [Advanced Physics Research 2.10: 2300044 \(2023\)](#)

Honors and Awards

- Travel Award to attend ELKIN 2026 [International Electrokinetics Society \(IES\), 2026](#)
- 2026 Winner of the APS-DBIO image contest [APS DBIO, 2026](#)
- Travel Award to attend APS Global Phsics Summit [Otto Mønsted Fonden, 2025](#)
- Negev Fellowship for Outstanding Graduate Students in Engineering (declined) [BGU, 2022](#)
- The Macquarie Group Scholarship from edX [edX, 2021](#)
- Institute Topper - IET PATW 2016 Presentation Competition “Microfluidics for the Future”
[Institution of Engineering and Technology \(IET\)- Present Around The World \(PATW\)](#)
- Invited Student Delegate at Indian Science Congress 2010

Invited Talks

- “Useful and Pointless Optima in Nature” [Biocomplexity Section, Niels Bohr Institute, Feb 2026](#)
- “Form v. Function through toy problems and actual toys” [Prakash Lab, Stanford, Nov 2025](#)
- “Traversing a thin film lubricant in finite time” [Alim Group, TU Munich, Aug 2024](#)

Conference Presentations

17. **J Sebastian**, H A Stone, K H Jensen. “Electrokinetic constraints on intercellular signalling in plants”
[ELKIN \(The Hague, NL – June 2026\)](#)

16. **J Sebastian**, K H Jensen. “*Electrokinetic constraints on intercellular signalling in plants*” APS DFD (Houston, USA – November 2025)
15. **J Sebastian**, K H Jensen. “*Geometric flows shaping universal geometries*” Complex Motion in Fluids – CMIF (Île d’Oléron, France – July 2025)
14. **J Sebastian**, K H Jensen. “*Random wear shapes all pointed things alike*” The Art and Science of Liquid Interfaces (Zurich, Switzerland – March 2025) Poster
13. **J Sebastian**, K H Jensen. “*Random wear shapes all pointed things alike*” APS Global Summit (LA, USA – March 2025)
12. **J Sebastian**, A L Schødt, K H Jensen. “*Traversing a thin film lubricant in finite times*” Complexity of Life Conference (Graz, Austria – September 2024) Poster
11. **J Sebastian**, A L Schødt, K H Jensen. “*Traversing a thin film lubricant in finite times*” Plant Biomechanics UK (Cambridge, UK – April 2024) Poster
10. **J Sebastian**, A L Schødt, K H Jensen. “*Traversing a thin film lubricant in finite times*” APS March Meeting 2024 (Minneapolis, USA – March 2024)
9. **J Sebastian** and Y Green. “*Can nanofluidic systems be described by a simple electrical circuit?*” 6th Physics of Membrane Processes Workshop - PMP 2023 (Online – November 2023)
8. **J Sebastian** and Y Green. “*Understanding the Conductance of Nanoslits*” Israel Society for Theoretical and Applied Mechanics, Annual Conference, (Technion, Israel – Dec 2022)
7. **J Sebastian** and Y Green*. “*Electrical Circuit Modelling of Nanofluidic Systems*” 75th Annual Meeting of the APS DFD 2022 (Indianapolis, USA – November 2022) (*Preseted by YG)
6. **J Sebastian** and Y Green. “*Electrical Circuit Modelling of Nanofluidic Systems*” 5th International Symposium on Physics of Membrane Processes - PMP 2022 (Wageningen, Netherlands – October 2022)
5. **J Sebastian** and Y Green. “*Electrical Circuit Modelling of Nanofluidic Systems*” 14th International Symposium on Electrokinetics - ELKIN 2022 (Tel Aviv, Israel – July 2022) Poster + Soundbite
4. **J Sebastian** and Y Green. “*Multichannel Nanofluidic Systems: The Equivalent Electrical Circuit*” International Water Summit (Sde Boker, Israel – May 2022)
3. **J Sebastian** and Y Green. “*Multichannel Nanofluidic Systems: The Equivalent Electrical Circuit*” 67th Annual Meeting of the Israel Physical Society (IPS) (Be’er Sheva, Israel – February 2022)
2. **J Sebastian** and Y Green. “*The Equivalent Electrical Circuit of Multichannel Nanofluidic Systems*” Israel Society for Theoretical and Applied Mechanics, Annual Conference, (Tel Aviv, Israel – December 2021)
1. **J Sebastian** and Y Green. “*Multichannel Nanofluidic Systems: The Equivalent Electrical Circuit*” Nano Israel 2021 (Jerusalem, Israel – October 2021)

Teaching and Mentoring

- Co-supervised 8 Msc students on research immersion projects (theory/experiments/numerics):
Fluid structure interactions, geometric erosion, microfluidics DTU: 2023 - 2026
- Teaching Assistance + Guest lectures (on transport, electrokinetics): DTU: 2023, 2024
Theoretical Microfluidics Graduate Course
- Teaching Assistance: *Statistical Physics* Graduate Course DTU: 2023, 2024
- Theory lectures + Lab tutorials: *Wind tunnel testing of airfoils* Short Course CET: 2017

Professional Research Experience

MRF Tyres (Research & Development)

July 2017 - September 2020

Engineer - Research; Executive - Research

Chennai, India

- Computational modelling of tyre composites; development of functional geometric designs and materials
- Image analysis methods to estimate complex deformations at the tyre contact patch

STAD TechnoInnovations

October 2015 - April 2016

Mechanical Design Intern

Ernakulam, India

- Reverse engineered the complex shape and material characteristics of the tibial bone implant to arrive at an internal mesh structure suitable for 3D printing

Leadership, Service, and Outreach

Organiser of weekly FLUIDS Friday Seminars at DTU Physics *2023 - 2026*

- Seminar invitations have led to collaborations with the section.

Panelist, Roundtable Discussion on Irreproducibility in Scientific Research *14 Mar 2022*

- INTERNATIONAL WORKSHOP: The Problem of Irreproducibility in Scientific Experimentation: Is there a “Replication Crisis”?

BGU, Israel

Member, History and Philosophy of Science Club

Nov 2021 - Mar 2022

- Presented analyses of historical texts at various meetings

BGU, Israel

Member, Emergency Design Collective (EDC)

Jun 2020 - Jul 2021

- Part of a global team of problem solvers working together to mitigate new challenges in the wake of the COVID-19 pandemic
- Nominated to top three projects under “Pandemic Era Educational Challenges”

Student Coordinator, Innovation Center CET

2016 - 17

- Coordinated weekly meetings, brainstorming sessions and training programs in the state-run facility
- Organised the first undergraduate thesis project expo, *Innov-EXPO* in May 2017

Bosch Student Ambassador

2016 - 17

- Coordinated research collaboration, projects and student placements

Founder and Host, CETALKS, an in- campus talk show at CET

2015 - 17

- Overall themes included ‘*Science in Art*’, and ‘*Engineering hidden in plain sight*’

Project Coordinator, Society of Automotive Engineers (SAE)

2015 - 16

- Launched and mentored eight projects and maximized participation in national design competitions